SIMATS ENGINEERING

ASSESSMENT TOOL I — Uncertainty Visualization Lab

Course: Data Handling and Visualization	**Code: DSA0601**	**CO: CO5**
Duration: 180 min	**Total Marks: 100**	**Reg No : 192424322** **Name:M.Arshini**

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

Code of Conduct:

I M.Arshini[192424322] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **M.Arshini**

Criteria	**Understanding of Uncertainty 25%**	**Visualization of Uncertainty 25%**	**Application of Visualization Principles 20%**	**Organization 20%**	**Accuracy 10%**	**Clarity 5%**	**Total**
Q1							
Q2							
Q3							
Q4							
Q5							

Uncertainty Visualization Lab – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualization, interpret uncertainty, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Bootstrapping and Confidence Intervals – Point Estimate

A quality-control team records the following delivery times (in minutes) for 12 randomly selected orders. Using R, calculate the sample mean and obtain a 95% bootstrap confidence interval for the mean using at least 5,000 bootstrap resamples. Visualize the bootstrap distribution and mark the observed mean.

Data: 31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36

Expected R tasks: resampling, point estimate, confidence interval,

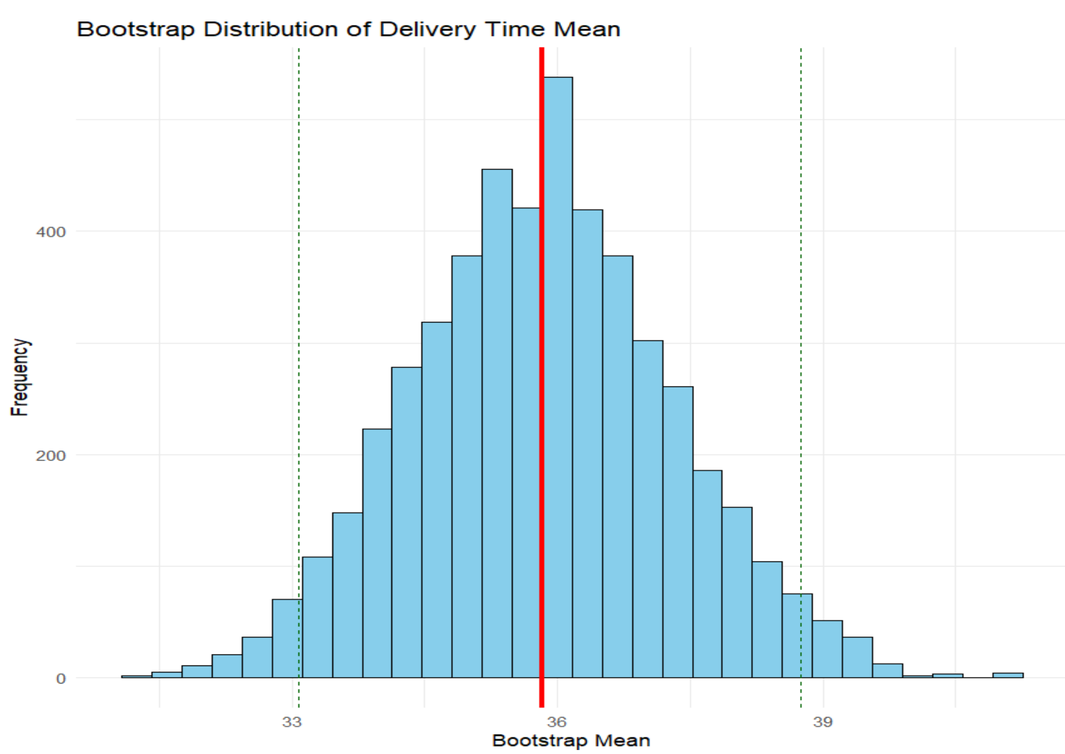
histogram/density plot, interpretation.

CODE:

```
library(ggplot2)

delivery_time <- c(31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36)

observed_mean <- mean(delivery_time)

set.seed(123)

n <- length(delivery_time)

B <- 5000

bootstrap_means <- replicate(B, {

  sample_data <- sample(delivery_time, size = n, replace = TRUE)

  mean(sample_data)

})

CI <- quantile(bootstrap_means, c(0.025, 0.975))

cat("Observed Mean =", observed_mean, "\n")

cat("95% Bootstrap CI =", CI[1], "to", CI[2], "\n")

bootstrap_df <- data.frame(mean = bootstrap_means)
```

```
ggplot(bootstrap_df, aes(x = mean)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  geom_vline(xintercept = observed_mean,
             color = "red", linewidth = 1.2) +
  geom_vline(xintercept = CI[1],
             linetype = "dashed", color = "darkgreen") +
  geom_vline(xintercept = CI[2],
             linetype = "dashed", color = "darkgreen") +
  labs(
    title = "Bootstrap Distribution of Delivery Time Mean",
    x = "Bootstrap Mean",
    y = "Frequency"
  ) +
theme_minimal()
```

OUTPUT:



INTERPRETATION

- The observed mean represents the average delivery time of the sample.
- The bootstrap distribution shows the variability of the estimated mean.
- The 95% confidence interval gives a plausible range for the population mean.
- The red line represents the observed sample mean.
- The dashed lines represent the lower and upper confidence limits.

Q2. Uncertainty in Trend Estimation

The following data represent monthly advertising expenditure and sales. Fit a linear regression model in R to estimate the sales trend. Plot the observations with the fitted regression line and a 95% confidence band. Explain how the uncertainty changes across the range of advertising expenditure.

Data: Advertising = 10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40;

Sales = 48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84

Expected R tasks: lm(), prediction interval/confidence interval, ggplot2

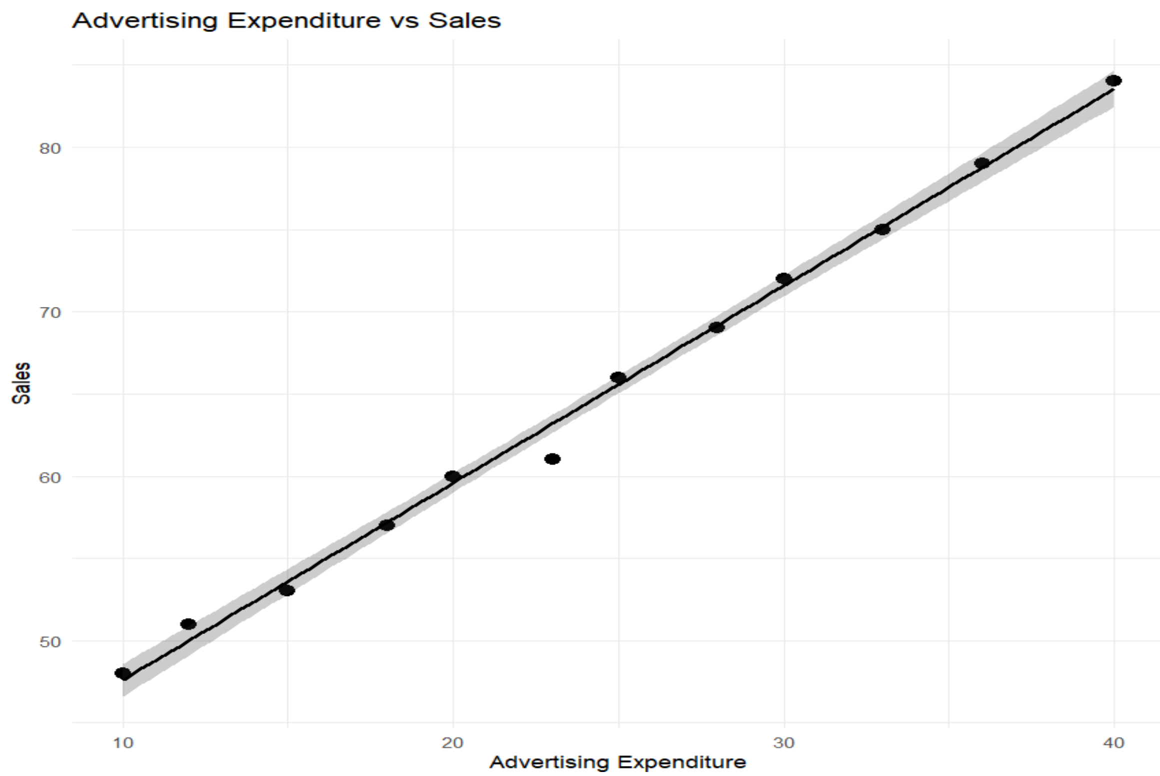
visualization, trend interpretation.

CODE:

```
library(ggplot2)

Advertising <- c(10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40)

Sales <- c(48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84)

data <- data.frame(Advertising, Sales)

model <- lm(Sales ~ Advertising, data = data)

summary(model)

new_data <- data.frame(

  Advertising = seq(min(Advertising),

                    max(Advertising),
```

```
                    length.out = 100)
)
pred <- predict(model, newdata = new_data,
                interval = "confidence")
prediction_data <- cbind(new_data, pred)
ggplot(data, aes(x = Advertising, y = Sales)) +
geom_point(size = 3) +
  geom_ribbon(
    data = prediction_data,
    aes(x = Advertising, ymin = lwr, ymax = upr),
    alpha = 0.25,
    inherit.aes = FALSE
  ) +
  geom_line(
    data = prediction_data,
    aes(x = Advertising, y = fit),
    linewidth = 1
  ) +
  labs(
    title = "Advertising Expenditure vs Sales",
    x = "Advertising Expenditure",
    y = "Sales"
  ) +
  theme_minimal()
```

OUTPUT:



INTERPRETATION:

- Sales increase as advertising expenditure increases.
- The regression line represents the estimated sales trend.
- The shaded region represents the 95% confidence band.
- The confidence band is generally narrower near the center of the observed advertising range.
- Uncertainty increases toward the extreme values because fewer observations support the estimated trend there.

Q3. Overplotting, Jittering and Transparency

A survey records the number of hours studied and examination scores for 30 students. Several students have identical values. Create a scatterplot in R first without modification, then redesign it using partial transparency and jittering. Compare the two plots and explain how overplotting affects interpretation.

Data: Hours = 2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,7,8,8,8,8,8,9,9,9,10,10;

Scores =

45,52,52,58,60,60,64,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94

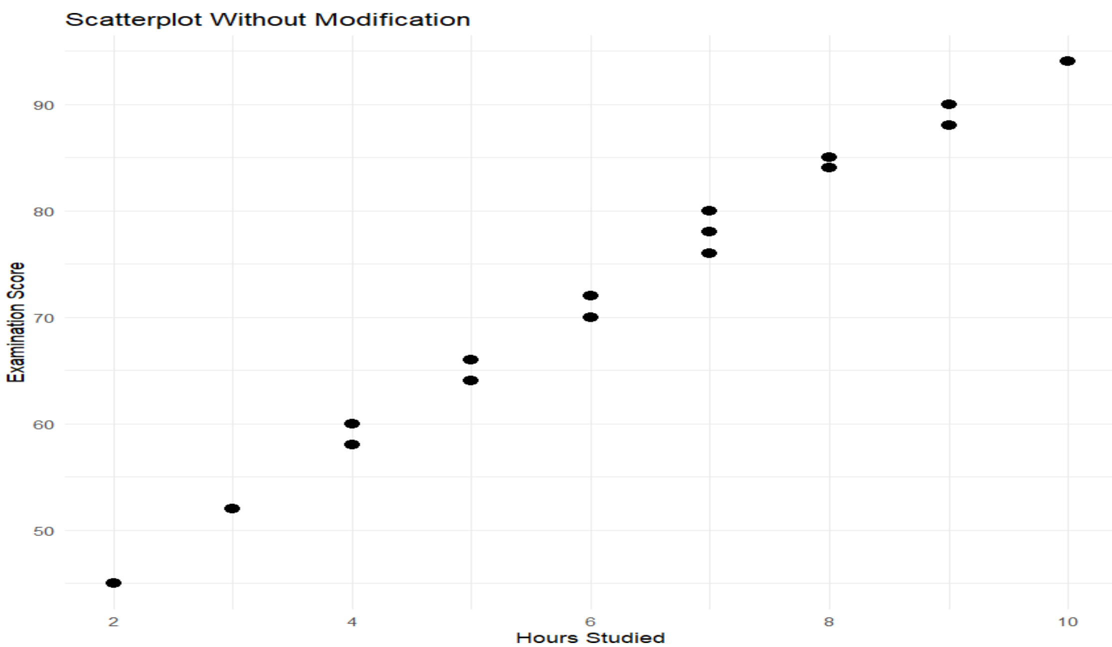
Expected R tasks: geom_point(), alpha, position_jitter(), comparison of overplotting.

CODE:

```
library(ggplot2)

Hours <- c(

2,3,3,4,4,4,5,5,5,5,

  6,6,6,6,6,7,7,7,7,7,

  8,8,8,8,8,9,9,9,10,10

)

Scores <- c(

  45,52,52,58,60,60,64,66,66,66,

  70,72,72,72,72,76,78,78,80,80,

  84,84,85,85,85,88,90,90,94,94

)

data <- data.frame(Hours, Scores)

ggplot(data, aes(x = Hours, y = Scores)) +

geom_point(size = 3) +

  labs(

    title = "Scatterplot Without Modification",

    x = "Hours Studied",

    y = "Examination Score"

  ) +

  theme_minimal()

ggplot(data, aes(x = Hours, y = Scores)) +

  geom_jitter(

    width = 0.15,

    height = 0.8,

    alpha = 0.5,

    size = 3

  ) +
```

```
labs(
  title = "Scatterplot with Jittering and Transparency",
  x = "Hours Studied",
  y = "Examination Score"
) +
theme_minimal()
```

OUTPUT:



INTERPRETATION:

- In the first plot, identical observations overlap.

- Overplotting can hide the actual number of observations.
- Jittering slightly moves points so overlapping observations become visible.
- Transparency allows dense regions to appear darker.
- The redesigned plot provides a clearer representation of the distribution of students.
- There is an overall positive relationship between study hours and examination scores.

Q4. 2D Histogram and Density Estimation

A researcher measures height and weight for 20 participants. Construct a 2D histogram and a 2D density visualization in R. Use the plots to identify regions with high observation density. State which visualization communicates concentration more clearly and justify your choice.

Data: Height = 150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186; Weight = 48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88

Expected R tasks: 2D binning, density estimation, contour/density layers,

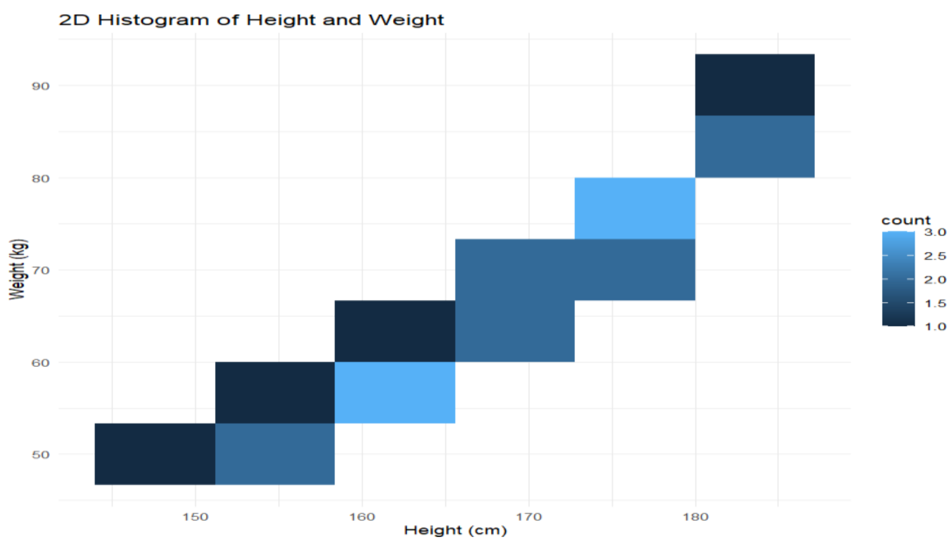
interpretation of concentration.

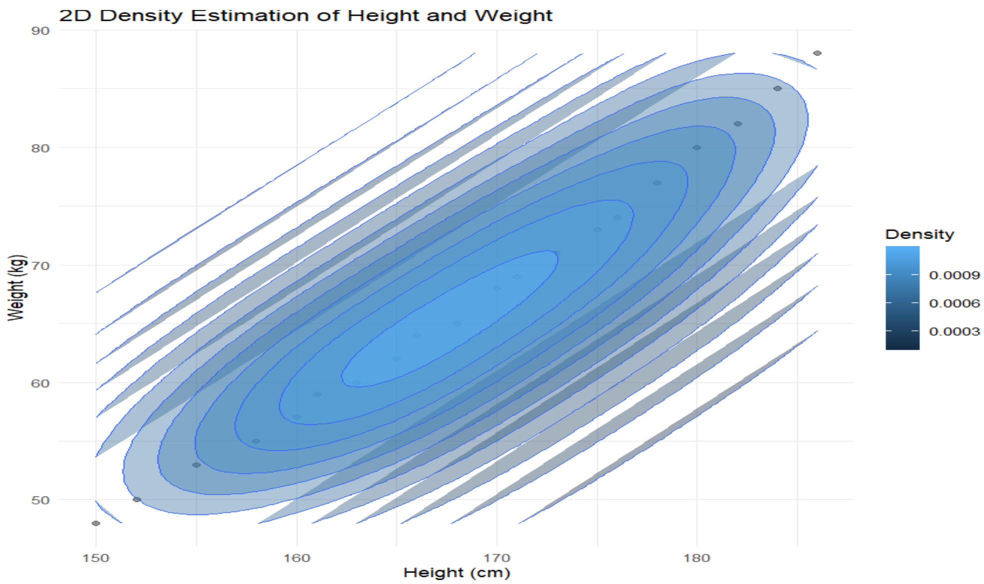
CODE:

```
library(ggplot2)

Height <- c(

  150,152,155,158,160,161,163,165,166,168,

  170,171,173,175,176,178,180,182,184,186

)

Weight <- c(

  48,50,53,55,57,59,60,62,64,65,

  68,69,71,73,74,77,80,82,85,88

)

data <- data.frame(Height, Weight)

ggplot(data, aes(x = Height, y = Weight)) +

geom_bin2d(bins = 6) +

  labs(

    title = "2D Histogram of Height and Weight",

    x = "Height (cm)",
```

```
    y = "Weight (kg)"
  ) +
  theme_minimal()
ggplot(data, aes(x = Height, y = Weight)) +
  geom_point(alpha = 0.4) +
  stat_density_2d(
    aes(fill = after_stat(level)),
    geom = "polygon",
    alpha = 0.4
  ) +
  geom_density_2d() +
  labs(
    title = "2D Density Estimation of Height and Weight",
    x = "Height (cm)",
    y = "Weight (kg)",
    fill = "Density"
  ) +
  theme_minimal()
```

OUTPUT





INTERPRETATION

- The 2D histogram divides the height-weight space into rectangular bins.
- Areas containing more observations represent higher concentration.
- The density plot provides a smoother representation of concentration.
- Most observations are concentrated along the increasing height-weight relationship.
- The 2D density plot communicates concentration more clearly because the smooth contours make high-density regions easier to identify.

Q5. Color Scales, Accessibility and Misleading Visualization

The following category-wise satisfaction scores must be visualized in R. Create two plots: (1) a plot using a nonmonotonic/diverging color scheme that could mislead interpretation, and (2) a redesigned plot using an ordered, color-vision-deficiency-friendly sequential scale. Explain why the second design is more effective and identify the visualization pitfall in the first design.

Data: Category = A,B,C,D,E,F,G;

Satisfaction = 42,48,55,61,67,74,82

Expected R tasks: color mapping, scale selection, accessibility-aware redesign, critique of misleading color use.

CODE:

library(ggplot2)

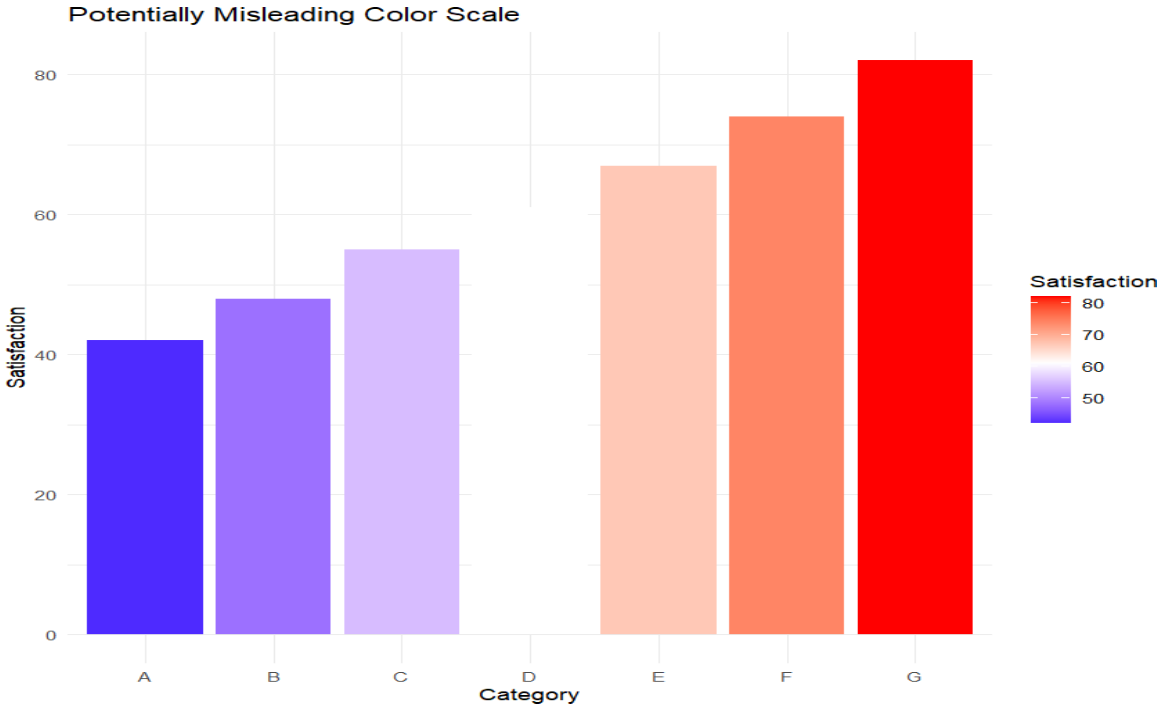
Category <- c("A", "B", "C", "D", "E", "F", "G")

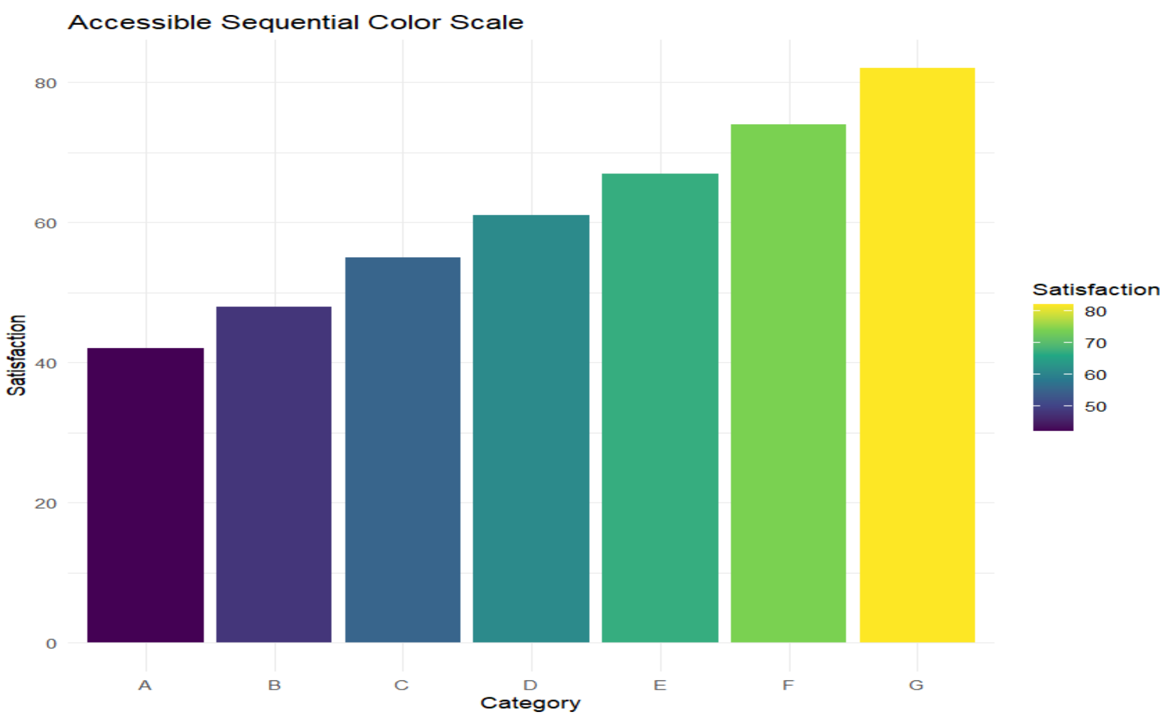
Satisfaction <- c(42, 48, 55, 61, 67, 74, 82)

data <- data.frame(Category, Satisfaction)

```
ggplot(data, aes(x = Category, y = Satisfaction, fill = Satisfaction)) +
  geom_col() +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 61
  ) +
  labs(
    title = "Potentially Misleading Color Scale",
    x = "Category",
    y = "Satisfaction",
    fill = "Satisfaction"
  ) +
  theme_minimal()
ggplot(data, aes(x = Category, y = Satisfaction, fill = Satisfaction)) +
  geom_col() +
  scale_fill_viridis_c(option = "viridis") +
  labs(
    title = "Accessible Sequential Color Scale",
    x = "Category",
    y = "Satisfaction",
    fill = "Satisfaction"
  ) +
  theme_minimal()
```

OUTPUT:





INTERPRETATION:

- The first plot uses a diverging color scale even though satisfaction is a single ordered variable.
- Red and blue can create an artificial visual distinction around the midpoint.
- This may make moderate values appear more important than they actually are.
- The second plot uses a sequential scale where color intensity follows the numerical ordering.
- The viridis scale is designed to be more accessible for people with color-vision deficiencies.
- Therefore, the second visualization communicates the satisfaction levels more accurately.